

# BANK LOAN RISK ANALYTICS

## Business Requirements & Answers — Consolidated Report

This report restates the project's business requirements as direct questions, each answered with a real figure produced by the corresponding tool. Each of the four sections below asks a different class of question so nothing is duplicated across the project: SQL answers portfolio-level performance, Python answers borrower-level risk and prediction, Excel answers point-of-need operational calculations, and Power BI answers ongoing executive monitoring.

### A. SQL (SQLite) — Portfolio Performance Requirements

Business need: understand the size, health, growth, and composition of the lending book as a whole.

| Business Question                                                      | Answer (from the analysis)                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How large is the loan portfolio, and how healthy is it overall?        | 38,576 total loan applications; \$435.8M funded against \$473.1M received; average interest rate 12.05%; average DTI 13.33%. Repayments already exceed funded principal at the portfolio level.                                                                                                                                                        |
| What share of the portfolio is performing vs. exposed to default risk? | 86.2% of applications (33,243 loans, \$370.2M funded) are Good Loans (Fully Paid or Current); 13.8% (5,333 loans, \$65.5M funded) are Bad Loans (Charged Off), of which only \$37.3M was recovered — a net shortfall of roughly \$28.2M.                                                                                                               |
| Which loan statuses contribute most to risk vs. revenue?               | Charged-Off loans carry the highest average interest rate (13.9%) and DTI (14.0%) of any status, ahead of Fully Paid (11.6% / 13.2%) — both variables run hotter for loans that ultimately default.                                                                                                                                                    |
| Is loan demand, funding, and repayment growing over time?              | Yes. Monthly funded amount rose steadily from \$25.0M in January to \$54.0M in December 2021 — roughly a 2.2x increase with no monthly declines, and repayments followed the same upward trend.                                                                                                                                                        |
| Which states, loan terms, and purposes matter most to volume and risk? | California (\$78.5M), New York (\$42.1M), Texas (\$31.2M) and Florida (\$30.0M) are the top four funding markets. 36-month loans make up ~73% of applications. Debt Consolidation is the largest purpose by funding (\$232.5M), while Small Business loans are disproportionately risky for their size (25.6% charge-off rate on only \$24.1M funded). |
| Which borrower segments repay most reliably?                           | Borrowers with 10+ years of employment tenure received the most funding (\$116.1M) and the strongest funded-to-received performance of any tenure band. Mortgage holders are the largest segment by volume and generate the highest amount received.                                                                                                   |
| Does the bank's own credit-grade system actually track real risk?      | Yes. Charge-off rate rises almost monotonically by grade — 5.7% for Grade A up to 31.3% for Grade G — and interest rate rises in lockstep, confirming the existing grading system is well-calibrated to real outcomes.                                                                                                                                 |

### B. Python — Borrower Risk & Predictive Modeling Requirements

Business need: understand which individual borrowers are likely to default and why, and whether that risk can be scored before a loan is funded.

| Business Question                                                                              | Answer (from the analysis)                                                                                                                                            |
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| Which loan purposes and borrower segments carry the most default risk at the individual level? | Small Business loans show one of the highest default rates of any purpose category. Charge-off likelihood also skews toward borrowers with shorter employment tenure. |

| Business Question                                                                                        | Answer (from the analysis)                                                                                                                                                                                                                                                                                                                                     |
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| Which loan and borrower attributes correlate most strongly with default?                                 | Interest rate shows the strongest positive correlation with default, followed by loan term and DTI; annual income is weakly negatively correlated. These three variables (rate, term, DTI) are the practical levers for risk-based pricing.                                                                                                                    |
| Can borrowers be usefully grouped into risk tiers using a simple score?                                  | Yes. A composite score of DTI + interest rate splits borrowers into equal-sized Low / Medium / High risk tiers with roughly double the default rate from the lowest to the highest tier — simple enough to replicate outside Python (see the Excel Risk Classification sheet).                                                                                 |
| Can loan default be predicted before funding, and how reliable is that prediction?                       | Yes, with meaningful lift over chance. A Logistic Regression model reaches ROC-AUC 0.687 (63.5% recall on actual defaults), and a Random Forest reaches ROC-AUC 0.684 with slightly higher accuracy (65.8% vs 63.5%). Because only 13.8% of loans default, accuracy alone is misleading — ROC-AUC and recall on the default class are the metrics that matter. |
| Which model should be used operationally, and why?                                                       | Logistic Regression is preferred for deployment: its ROC-AUC is statistically comparable to the Random Forest, but its coefficients are fully transparent and explainable — a material advantage for underwriting and fair-lending compliance review, where a 'black box' score is harder to justify to regulators or applicants.                              |
| Which two variables should underwriting prioritize first, and is that confirmed by more than one method? | Interest rate and DTI. Both the correlation analysis and the feature-importance rankings of two independently trained models (Logistic Regression and Random Forest) agree these are the top two drivers of default — this is not one method's opinion, it is a convergent finding.                                                                            |

## C. Excel — Point-of-Need Operational Requirements

Business need: give front-line staff (loan officers, risk analysts, reporting teams) instant, formula-driven answers for a single applicant or a quick check, without opening a database or a Python environment.

| Business Question                                                                                                                                                                      | Answer (from the analysis)                                                                                                                                                                                                                                                                           |
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| If a loan officer needs the exact monthly payment and total interest cost for one applicant's proposed loan, on the spot, without opening Python or a database — how is that computed? | The Loan Amortization sheet uses PMT / IPMT / PPMT / NPER / CUMIPMT formulas driven by three editable inputs (amount, rate, term). Example shown: a \$10,000 / 11.76% / 36-month loan carries a \$331.00 monthly payment and \$1,915.93 total interest — recalculated instantly for any new figures. |
| How can a risk analyst triage a new application into Low / Medium / High risk without access to the Python model?                                                                      | The Risk Classification sheet reproduces the same tiering logic with two INDEX/MATCH band lookups (DTI band, Rate band) feeding a nested risk-tier matrix — no code required, and the bands are editable in one place if underwriting policy changes.                                                |
| Does the simple average interest rate on charged-off loans understate the true dollar-weighted cost of those defaults?                                                                 | Yes. The loan-size-weighted average rate on Charged-Off loans (SUMPRODUCT) is 14.71%, noticeably higher than the simple, unweighted average of 13.88% (AVERAGEIFS) — meaning the larger charged-off loans tend to carry higher rates than smaller ones, compounding their dollar impact on losses.   |
| How can top-market performance be communicated in one readable line for a report or email, not a table a reader has to interpret?                                                      | TEXTJOIN formulas on the Advanced Lookups sheet auto-build a plain-language summary per state, e.g. 'CA: 6,894 loans   \$78,484,125 funded', refreshed automatically if the underlying data changes.                                                                                                 |

| Business Question                                                                                                                     | Answer (from the analysis)                                                                                                                                                                                                               |
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| Which employment-length segments can be compared on a received/funded ratio without the formula breaking if a segment has zero loans? | An IFERROR-wrapped ratio table on the Advanced Lookups sheet computes Received ÷ Funded for every employment-length band and returns 'n/a' instead of a #DIV/0! error for any empty segment, keeping the table safe to filter or extend. |

## D. Power BI — Executive Monitoring Requirements

Business need: give leadership a live, drillable view of the same portfolio for ongoing monitoring, distinct from the one-time analysis done in SQL, Python, and Excel.

| Business Question                                                                                            | Answer (from the analysis)                                                                                                                                                                                                                                                           |
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| How does leadership monitor these KPIs on an ongoing basis, not just as a one-time report?                   | The Power BI Summary page shows Total Applications, Funded Amount, Amount Received, Average Interest Rate, and Average DTI, each with live MTD and MoM DAX measures, so month-over-month drift (e.g. DTI rising from 13.33% to 13.67% MTD) is visible without rebuilding the report. |
| Where can a manager drill from a portfolio-level number down to the individual loans behind it?              | The Details page lists transaction-level records (ID, purpose, home ownership, grade, sub-grade, issue date, funded amount, rate, installment, amount received) with State/Grade/Purpose slicers, so any KPI on the Summary or Overview pages can be traced to its underlying loans. |
| Which visuals let the bank compare growth and risk across time, geography, and borrower segment at a glance? | The Overview page combines a monthly trend line (funding and receipts), a U.S. state map, and breakdowns by term, employment length, purpose, and home ownership on one screen, so growth and concentration can be read side by side.                                                |

## Consolidated View

Read together, the four sections form a single chain rather than four separate analyses: SQL establishes that the portfolio is healthy but that 13.8% of funded principal turns into a \$28.2M net loss; Python explains that this loss is concentrated in specific, predictable borrower profiles (high rate, high DTI, Small Business purpose) well before those loans are funded; Excel puts that same classification logic into a form a non-technical user can apply to one applicant at a time; and Power BI keeps the whole picture visible to management every month so the effect of any underwriting change can be tracked going forward.